

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 7 — Electrochemistry**Total Marks: 65****Time Allowed: 2h 30m****Version: 1****General Instructions:**

1. Write your name, roll number and section clearly on the answer sheet.
2. Section A is compulsory. Encircle the correct option on the question paper itself.
3. In Section B, attempt **all 11** questions. For each question, attempt either the main question **OR** its alternative.
4. In Section C, attempt **all 4** questions. For each question, attempt either the main question **OR** its alternative.
5. Use black or blue ink only. Pencil is not allowed except for diagrams.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	In which change is nitrogen <i>reduced</i> ?	N ₂ to NO	N ₂ to NO ₂	N ₂ to NH ₃	N ₂ to HNO ₃	ⒶⒷⒸⒹ
2	Which is an example of <i>oxidation</i> ?	Cl ₂ → Cl ⁻	Ag → Ag ⁺	O ₂ → O ²⁻	Fe ³⁺ → Fe ²⁺	ⒶⒷⒸⒹ
3	In ZnO + H ₂ → Zn + H ₂ O, which substance is reduced?	H ₂	ZnO	Zn	O	ⒶⒷⒸⒹ
4	Oxidation state of Cr in K ₂ Cr ₂ O ₇ is:	+12	+6	+3	-6	ⒶⒷⒸⒹ
5	Oxidation state of S in SO ₄ ²⁻ is:	+2	+4	+6	-2	ⒶⒷⒸⒹ
6	In H ₂ S + Cl ₂ → 2HCl + S, H ₂ S acts as:	Reducing agent	Oxidizing agent	Catalyst	Electrolyte	ⒶⒷⒸⒹ
7	An oxidizing agent is a substance that:	loses electrons	gains electrons	loses oxygen	gains hydrogen	ⒶⒷⒸⒹ
8	Iron(III) chloride (IUPAC name) has the formula:	FeCl	FeCl ₂	FeCl ₃	Fe ₂ Cl ₃	ⒶⒷⒸⒹ
9	Rusting of iron requires:	oxygen only	water only	oxygen and water	CO ₂	ⒶⒷⒸⒹ
10	Cathodic (sacrificial) protection of ships uses blocks of:	copper	tin	zinc	magnesium	ⒶⒷⒸⒹ
11	Acidified KMnO ₄ changes from purple to _____ when a reducing agent is added:	blue	colourless	reddish-brown	yellow	ⒶⒷⒸⒹ
12	The oxidation number of any free element (e.g. Zn, H ₂) is always:	+1	-1	zero	+2	ⒶⒷⒸⒹ

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt all 11 questions. For each pair, attempt either the main question OR its alternative.

Part	Question	Marks	OR	OR Question	Marks
2(i)	Define oxidation and reduction in terms of (a) oxygen and (b) electrons.	3	OR	Define oxidation and reduction in terms of hydrogen transfer. Give one example of each.	3
2(ii)	State the four rules for assigning oxidation states. Use rule 4 to find the oxidation state of N in HNO ₃ .	3	OR	What is the oxidation state of a free element? What is the oxidation state of a monoatomic ion? Find the oxidation state of Na in Na ⁺ and Fe in Fe ³⁺ .	3
2(iii)	Determine the oxidation state of Cr in K ₂ Cr ₂ O ₇ . Show all working.	3	OR	Determine the oxidation state of C in carbonate ion CO ₃ ²⁻ . Show all working.	3
2(iv)	Find the oxidation state of N in: (a) NO ₂ (b) N ₂ O (c) HNO ₃ .	3	OR	Find the oxidation state of S in: (a) H ₂ S (b) SO ₂ (c) Na ₂ S ₂ O ₃ .	3
2(v)	Write the formula of: (a) copper(II) sulphate (b) iron(II) chloride (c) iron(III) chloride. State the oxidation state of the metal in each.	3	OR	Derive the formula of: (a) calcium chloride (Ca ²⁺ , Cl ⁻) (b) aluminium sulphate (Al ³⁺ , SO ₄ ²⁻). Show the charge-balancing step.	3
2(vi)	Define oxidizing agent and reducing agent. Name three common oxidizing agents and two common reducing agents.	3	OR	In 2Na + Cl ₂ → 2NaCl, identify the oxidizing and reducing agents. Justify using oxidation states.	3
2(vii)	In Fe ₂ O ₃ + 3CO → 2Fe + 3CO ₂ , identify: (a) element oxidized (b) element reduced (c)	3	OR	In WO ₃ + 3H ₂ → W + 3H ₂ O, identify: (a) substance oxidized (b) substance reduced (c)	3

	oxidizing agent (d) reducing agent.			oxidizing agent (d) reducing agent.	
2(viii)	Describe Activity 7.1: what happens when acidified KMnO_4 is added to FeSO_4 ? State the oxidizing and reducing agents.	3	OR	Describe Activity 7.2: what colour change occurs when H_2O_2 is added to KI solution? Write the equation and name the oxidizing agent.	3
2(ix)	Define corrosion. Write the two half-reactions that occur during rusting of iron, stating which is oxidation and which is reduction.	3	OR	Why are both oxygen and water necessary for iron to rust? What is the chemical formula of rust?	3
2(x)	Describe galvanizing and electroplating as methods of preventing corrosion. How does each work?	3	OR	Explain cathodic (sacrificial) protection. Why are magnesium blocks used on ships? Why must the sacrificial metal be more active than iron?	3
2(xi)	Find the oxidation state of the central atom in: (a) SO_4^{2-} (b) PO_4^{3-} (c) NH_4^+ . Show working.	3	OR	Why are Roman numerals used in certain compound names? Write the IUPAC names of FeCl_2 , FeCl_3 , and CuSO_4 .	3

SECTION C — Long Questions (4 × 5 = 20 Marks) — Attempt all 4

Part	Question	Marks	OR	OR Question	Marks
3(i)	(a) Define redox reactions. Explain oxidation and reduction in terms of all three definitions: oxygen, hydrogen, and electron transfer. [3] (b) In $\text{CuO} + \text{Zn} \rightarrow \text{Cu} + \text{ZnO}$, rewrite as an ionic equation and use oxidation states to show which species is oxidized and which is reduced. [2]	5	OR	(a) State all four rules for assigning oxidation states. Explain Rule 3 (complex ion) with an example. [3] (b) Find the oxidation state of: (i) B in H_3BO_3 (ii) N in NH_4^+ (iii) Mn in KMnO_4 . Show full working. [2]	5
3(ii)	(a) Define oxidizing and reducing agents. Explain why an oxidizing agent is itself reduced and a reducing agent is itself oxidized. [2] (b) Identify the oxidizing and reducing agents in: (i) $2\text{S} + \text{Cl}_2 \rightarrow \text{S}_2\text{Cl}_2$ (ii) $2\text{Na} + \text{Br}_2 \rightarrow 2\text{NaBr}$ (iii) $\text{Mg} + \text{H}_2\text{O} \rightarrow \text{MgO} + \text{H}_2$. Justify using oxidation states. [3]	5	OR	(a) What colour change confirms a reducing agent when using acidified KMnO_4 ? State the change in oxidation number of Mn and what it indicates. [2] (b) Identify the oxidizing and reducing agents in: (i) $2\text{KI} + \text{Cl}_2 \rightarrow 2\text{KCl} + \text{I}_2$ (ii) $2\text{FeCl}_2 + \text{Cl}_2 \rightarrow 2\text{FeCl}_3$ (iii) $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$. [3]	5
3(iii)	(a) Describe the three steps to derive the formula of an ionic compound. Apply them to write the formula of aluminium sulphate (Al^{3+} , SO_4^{2-}). [3] (b) Write the formula and state the metal's oxidation number for: (i) copper(I) oxide (ii) copper(II) oxide (iii) iron(II) sulphate (iv) manganese(IV) oxide. [2]	5	OR	(a) What are Roman numerals used for in chemical names? Give two examples from the textbook. [2] (b) Determine the oxidation state of the underlined element: (i) N in NO_2 (ii) S in H_2SO_4 (iii) Cr in Cr_2O_3 (iv) Mn in MnO_4^- . Show full working. [3]	5
3(iv)	(a) Define corrosion. Explain why it is an electrochemical process. Write and label the anodic and cathodic half-reactions for the rusting of iron. [3] (b) Describe any four methods of preventing corrosion and state the principle behind each. [2]	5	OR	(a) Explain cathodic (sacrificial) protection in detail. Why is magnesium preferred over copper? Name two structures protected this way. [3] (b) Compare tinning and galvanizing under: (i) metal used (ii) process (iii) typical application. [2]	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Chapter 7 — Electrochemistry (pp. 113-126)

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